

Lab 4: Forecasting with decomposition

MATH840 | Time Series | Week 4 — graded assignment 3 of 7

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! What changed this week

You asked to do the graded work at home, and that is how it works from now on.

- **The practice session is no longer the lab.** We spend it on worked examples and discussion — the moves you are about to need, on a shared dataset, with everything asked out loud.
- **The graded work is done at home, due 23:59 on the day of the practice session.** No in-class submission, no baseline deadline.
- **Attendance is not marked.** The session is worth coming to because the examples are the ones you will reuse an hour later, not because a register is taken.

💡 Open the notebook

 [Open in Colab](#)

[Open In Colab](#)

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

🎯 The task

Today you are issued **your own time series**, and it stays yours for the rest of the course.

This lab asks you to take it apart and forecast it from its parts: decompose it, forecast the seasonally adjusted series, carry the seasonal component forward, put them back together — and then compare the result honestly against the four simple methods from Week 1. Toolbox only: no ETS, no ARIMA. Those are Weeks 6 and 7, on this same series.

! This one counts

8 points. **Deadline: 23:59 today**, on Moodle. Marks and comments come back before the next practice session.

📁 Your series

Your series is issued by a hash of the **same identifier you have used since Week 1**. The template does it for you: it turns your identifier into a code such as `TS-4DD7A` and downloads that series.

i What “anonymised” means here

The series is real. What has been removed is its identity: no name, no units, values rescaled, the calendar shifted by whole years, the history trimmed. Two students never get the same window of the same series, so a forecast is worth exactly what your model is worth.

You are not expected to work out what it measures, and you will not be marked for trying. Read the data, not the domain.

This series is yours for Weeks 6, 7, 8 and 9 as well. Every series in the pool is seasonal, has at least fifteen years of history, and has been checked so that both an automatic ETS and an automatic ARIMA beat its benchmark — which is what makes the challenges from Week 6 a fair fight.

What you are marked on

Section	Criteria	Points
1	EDA and visualisation	1
2	Decomposition and forecast	3
3	Justification block	3
4	Code quality and reproducibility	1
Total		8

The notebook has one section per row. **Keep those headings.**

Why accuracy is not marked this week — and is from Week 6

The plan was to mark this week on whether you beat your benchmark. Before setting that, the odds were measured on the actual pool, and they do not support it:

Benchmark	Decomposition forecast beats it
the best of the four simple methods, chosen on a validation window	36%
a plain naïve forecast	96%

A target you clear 36% of the time no matter how well you work is luck, and one you clear 96% of the time is a formality. Neither is worth three points. So this week the points go to the work itself — how the decomposition is done, and how well you argue from it — and **accuracy is measured but not marked**: your score against all four simple methods is published the next morning as information.

From Week 6 that changes. With ETS, and then with ARIMA, every series in the pool is one where the method of that week does beat the benchmark — so the accuracy points start when they measure your model rather than your draw. This week's number is the baseline they will be compared against.

Section 1 — EDA and visualisation (1 point)

This is a series you have never seen. The plots are here to justify the decisions in Section 2, not to fill space.

- A time plot, always.
- Whichever of the seasonal, subseries and ACF plots your argument actually uses.
- The validation split on a plot: history, validation window, and your forecast over it.
- One sentence per plot. **A plot with no reading attached earns nothing.**

Section 2 — Decomposition and forecast (3 points)

Item	Points	What earns them
2.1	1	The decomposition. STL with period equal to your series' seasonal period, components plotted, and the seasonally adjusted series computed from it. A decomposition with the wrong period earns nothing here, and nothing after it.
2.2	1	The forecast, built from the parts. The seasonally adjusted series forecast with a simple method, the seasonal component carried forward, the two added back. Candidates compared on the validation window — the last h observations — with MASE, against all four simple methods.
2.3	1	The submission. Your chosen approach refit on the whole history, h steps ahead, with an 80% interval, written to the submission file by the notebook.

⚠ Where this goes wrong

- The seasonal period is `M` from the setup cell — 12 for monthly series, 4 for quarterly. Not the number of years, not 7.
- Forecast the **adjusted** series, not the raw one, and add the season back afterwards. Doing one without the other is the most common way to lose both marks in 2.2.
- `STL` needs `seasonal` to be an odd integer ≥ 3 , and an `int`.
- Refit on the whole history before forecasting the hidden horizon. Forecasting from the training part only throws away your last h observations — the most informative ones you have.

📖 [FPP 5.7 forecasting with decomposition](#) · [FPP 5.4 residual diagnostics](#) · [FPP 5.8 evaluating accuracy](#) · [FPP the Pythonic Way, ch. 5](#)

Your benchmark, for comparison

The template computes it: the **best of the four simple methods** — `mean`, `naive`, `snaive`, `drift` — chosen by RMSE on the validation window, then refit on the whole history. It is not a target you are marked against this week, but it is the number Section 3 argues with, and the one the published score will use.

📁 Section 3 — Justification block (3 points)

Three points, and the part a human reads closely. Roughly a paragraph each, with numbers from your own cells rather than adjectives.

Item	Points	What earns them
3.1	0.75	What the data shows. Frequency, seasonal period, trend, level shifts, outliers — each claim tied to a plot from Section 1.
3.2	0.75	Why this decomposition and this level method. Your choice of <code>seasonal</code> and <code>robust</code> , and of the method that forecasts the adjusted series, tied to what you just described. Say what you tried and dropped.

Item	Points	What earns them
3.3	0.75	What the residuals say. ACF and a Ljung-Box test on your final model's residuals, read: what structure is left, and what it damages — the point forecast or the interval.
3.4	0.75	What you expect. Your MASE on the validation window against all four simple methods, and what you expect on the hidden horizon. If you expect to lose to a simple method, say which and why.

⚠ Important

Marked on the reasoning. A decomposition forecast that loses to seasonal naive and explains exactly why scores full marks here; one that wins and cannot say why does not.

🔪 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: nothing after the end of your history exists, so nothing can be fitted to it. If you tuned on the validation window, say so in 3.2 — that is legitimate, and it is also why your validation MASE is optimistic.
- Every number in your text comes from a cell.

📁 Submission

Upload three files to the Week 4 activity on Moodle **before 23:59 today**:

- `SURNAME_lab04_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
<code>student_id</code>	your identifier

Column	What
<code>code</code>	your series code, e.g. <code>TS-4DD7A</code>
<code>ds</code>	the dates you are forecasting, in order
<code>yhat</code>	your point forecast
<code>yhat_lo_80</code> , <code>yhat_hi_80</code>	your 80% interval

- `SURNAME_lab04.pdf` — the report, with the justification block;
- `SURNAME_lab04.ipynb` — the source.

The format matters even though accuracy does not

Item 2.3 is the submission file, and it is marked on being correct: exactly h rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one used to read your file, so if it passes, your file is fine.

How long this should take

Ninety minutes if you follow the practice session's example, half of that if you reuse its code.

For next week

Week 5 is the mid-term quiz — no practice session and no graded work. It covers Weeks 1–4: forecasting basics and graphics, decomposition, the toolbox, exponential smoothing.

Week 6 is Challenge 2, on this same series, with ETS — the Week 4 lecture, already published. That one is marked on accuracy against your benchmark, and today's number is what it will be compared against. Read [FPP3 ch. 8](#) before then and your challenge is half done.